Malaria & Dengue Early Warning for Pakistani Districts - Summary

User & decision: district health officers (Sindh, Balochistan, KP) deciding where to send RDT kits, antimalarials, beds and fumigation teams next week. **Output:** next-week suspected cases per district with an 80% range + cited NIH/WHO guidance in English and Urdu.

Data & quality

NIH IDSR bulletins (177 PDFs, 203,895 cells), Open-Meteo hourly weather (2,905,770 rows), OpenDengue, HDX. Raw -> cleaned -> gold in DuckDB; `make data` rebuilds from empty folders and verifies SHA-256 for 259 raw files. The Data Quality Agent logged FILL 1, FIX 6, FLAG 16, PASS 8 decisions in the last run (e.g. 'NR' and 'not in top-10' kept missing, not zero; short weather gaps filled and marked; high-risk fixes need human approval).

Model vs declared baseline (test period scored once)

	Baseline: last value	LightGBM quantile (median)
MAE (suspected cases / district-week)	117.0	110.1
RMSE	273.9	254.0
MAE improvement (95% block bootstrap)		6.0% (-0.3% to 12.0%)
80% interval coverage		78.0%

Test: 4,274 district-weeks, 97 districts, target weeks 2025-10-06 to 2026-09-21. Smallest gain over the baseline: Khyber Pakhtunkhwa (MAE 30.44 vs baseline 31.14); full tables by province, poverty tier, season and district on the dashboard.

Insight Agent (RAG)

Hybrid retrieval over NIH advisories/guidelines, WHO fact sheets and the latest bulletins; numbers only from gold SQL with row ids. Golden set (17 questions, EN+UR): refusal accuracy 100%, citation accuracy 100%, number faithfulness 100% (22 numbers re-read by id).

Automation

Prefect flow every hour (cron 5 * * * *): fetch -> Gate 1 -> Gate 2 (freshness, drift) -> retrain only on new weeks/drift (keep if better) -> re-index -> golden eval -> publish to a Hugging Face Space showing freshness, quality status and last successful run.

Business model & Pakistan impact

Free to districts, funded by provincial health departments or malaria/dengue programme partners (contract/grant); small running cost (one small VM, free Space tier for a pilot, optional LLM wording at about $0.01 per answer). Built entirely on Pakistani surveillance data, with Urdu outputs. Local validation: our parsed NIH national dengue totals equal OpenDengue's monthly Pakistan series in 15 of 20 months (r = 0.9973); 4 other months miss a bulletin week, 1 differs unexplained. District sums are checked against each bulletin's province total.

Limits: suspected cases; Punjab/ICT districts not in tables; no gender breakdown; 1-2 week publication lag. Information, not medical advice - emergencies: nearest hospital / Rescue 1122.